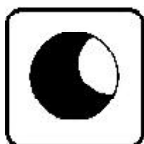


Operator's Manual

8 Channel High Resolution Brain Array for the GE 1.5T Signa Infinity with Excite Technology



Manufactured by:

MRI Devices Corporation
1515 Paramount Drive, Suite A
Waukesha, Wisconsin 53186 USA

Distributed by:

GE Medical Systems



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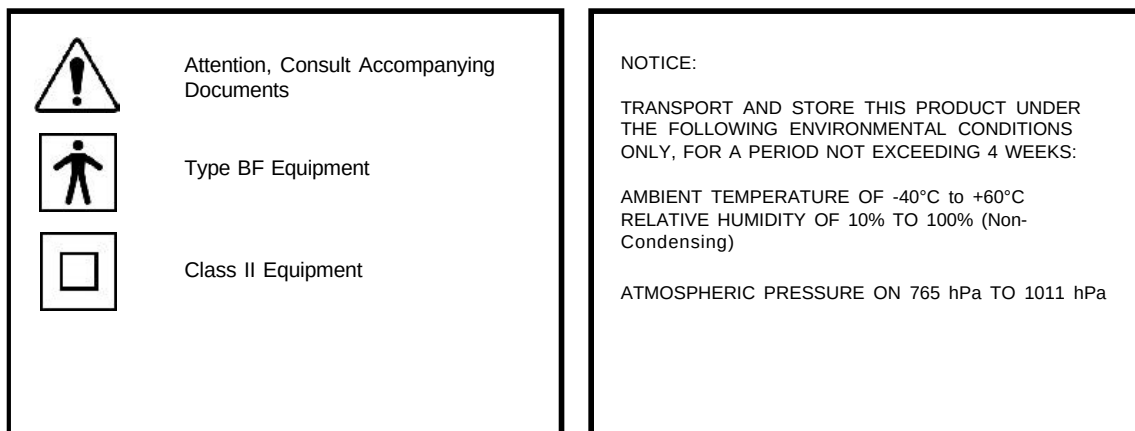
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"Signa Infinity with Excite Technology" is a registered trademark of the General Electric Company.

Proper performance of this coil is guaranteed only while the coil is being used on the MR system (hardware/software level) specified at the time of purchase. Upgrades or other modifications to the system software and/or hardware may affect compatibility. Prior to upgrading your MR system, please contact your GE Medical Systems representative to discuss coil compatibility issues. Failure to do so may void your warranty.

Medical Device Directive

These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:



Caution:
Federal law restricts this device to sale,
distribution, and use by or on the order of
a physician.

Introduction

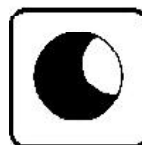
This manual describes the safety precautions, features, use and care of the MRI Devices 8 Channel High Resolution Brain Array, compatible with the GE 1.5T Signa Infinity with Excite Technology. Please review this manual thoroughly before using the device.

If you have any questions or comments on this manual, or need any assistance with the use of the product, please contact your GE Medical Systems representative at:

1-800-GE CARES

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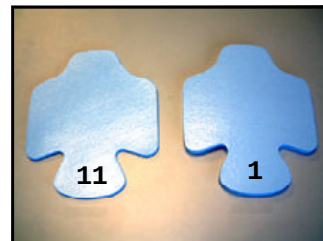
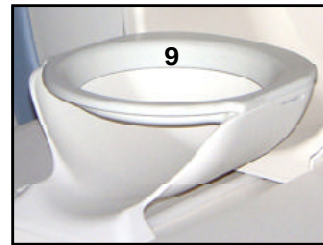


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8 Channel High Resolution Brain Array

The 8 Channel High Resolution Brain Array, consists of the following parts. Please inspect upon receipt to make sure all parts have arrived and are in good order. Use this guide to refer to part names throughout this manual.



- 1. Thick Head Pad
- 2. Head Holder
- 3. Mirror
- 4. Window
- 5. Base
- 6. Locking Lever
- 7. Coil Cable
- 8. Coil Connector
- 9. Phantom Holder
- 10. Wedge Pads
- 11. Thin Head Pad

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Chapter 1 - Safety

Training

This manual contains detailed information on the set-up, positioning and use of your MRI Devices Corporation coil. The instructions should be read carefully and thoroughly before attempting to scan patients with the coil.

Quality Assurance

The procedure described in the Quality Assurance Section of this manual should be performed upon receipt of the coil to establish a baseline of coil performance. The procedure should be repeated at regular intervals.

Indications

The coil is indicated for use, on the order of a physician, in conjunction with an MR scanner as an accessory to produce general purpose images.

Contra-indications

The operator should be aware of the following contraindications for use related to the strong magnetic field of the MR system:

 Scanning is contraindicated for patients who have electrically, magnetically or mechanically activated implants (for example, cardiac pacemakers). The magnetic and electromagnetic fields produced by the MR System and coil may interfere with the operations of these devices.

 Scanning patients with intracranial aneurysm clips is contraindicated.

Precautions

Precautions should be taken when scanning patients with the following conditions:

 Greater than normal potential for cardiac arrest

 An increased likelihood for developing seizures or claustrophobia

 Unconscious, heavily sedated, or confused patients

 Patients with whom reliable communications cannot be maintained

Cautions

The following general warning statements apply to scanning with a magnetic resonance system. For further details, review the warnings in your MR system Operators Manual.

-  Cables should not be looped or crossed. Arcing and patient burns could result.
-  If a patient complains of burning, tingling, stinging, or other similar 'burn'-type sensations, promptly stop the scan procedure, examine the patient, and contact the responsible physician before continuing the procedure. Pay special attention to very young, sedated, or other compromised patients who may not be able to communicate effectively.
-  Route the cables out of the magnet so that they do not touch the patient.
-  Patients with ferromagnetic metal should not be scanned because the magnetic field may interact with implanted surgical clips or other ferromagnetic materials.
-  Persons with cardiac pacemakers or other implanted electronic devices should not enter the magnetic field zone delineated by the MR system manufacturer.
-  There is a risk to scanning feverish or decompensated cardiac patients.
-  Facial makeup should be removed before scanning because it may contain metal flakes which can cause skin and eye irritation. Permanent eyeliner tattoos may cause eye irritation due to ferromagnetic particles.
-  Patients who work in environments in which there is a risk of having embedded metallic fragments in or near the eye should be carefully screened before undergoing an MR exam.
-  Visually inspect the white insulator jacket and the coil connector with each use. If the insulator is broken, or the cable is frayed at the connector, immediately discontinue use of the device.



Emergency Procedures

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise, or if the patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.

Chapter 2 - Quality Assurance

SNR Protocol

This section contains the Coil Imaging Performance Verification procedure from the GE 1.5T 8 Channel High Resolution Brain Array Service Manual. The information is duplicated complete with Service Manual section subheadings and numbering.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	MRIDC #	Qty
18cm Diameter Spherical TLT Phantom	46-265826G6	N/A	1
Phantom Positioner	2317112-3	102558	1

3-2-2 Data Collection

1. Attach the base plate with 8 channel hi res brain coil to the cradle in the same position on the cradle as the standard head coil. Note the serial number of the coil.
2. Slide the coil back to permit access to the head holder as shown in Figure 3-2-2-1.
3. Place phantom ring on head holder as shown in Figure 3-2-2-2. The edge of the ring that goes toward the magnet has edges that overhang the head holder to secure the ring in position.
4. Place head TLT sphere on phantom ring as shown in Figure 3-2-2-3.
5. Slide the coil all the way forward (away from the magnet) on the base plate and set the locks as shown in Figure 3-2-2-4.
6. Establish an axial landmark on the line on top of the coil. Advance to scan.
7. End the previous exam if this has not already been done.
8. On the service desktop, open a command window and navigate to the **/usr/local/bin/CoilsTools** directory.
9. Enter the command **run8hrbrainQA** and press the enter key.
10. When scanning is completed and images have been extracted (messages stop scrolling in the command window and the prompt returns, approximately 2 ½ minutes), note the exam number and continue with the SNR Analysis section.
11. The images will appear similar to those seen in Figure 3-2-2-5 (ROI's will not be shown).

Figure 3-2-2-1



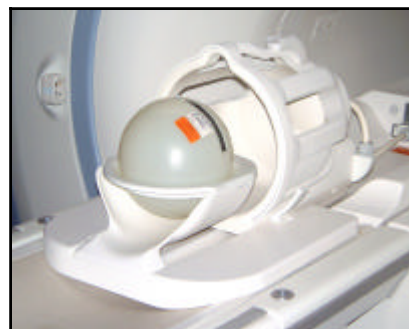
Slide Coil Back to Access Head Holder

Figure 3-2-2-2



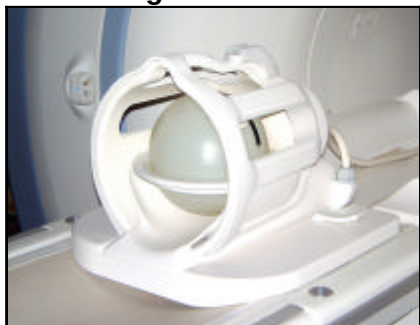
Place Phantom Ring on Head Holder

Figure 3-2-2-3



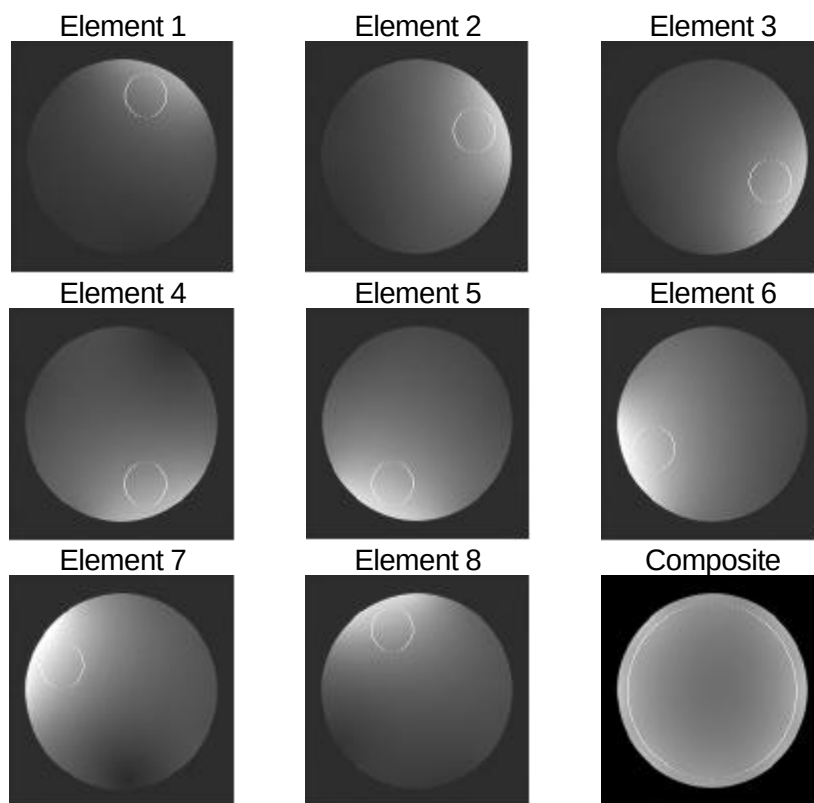
Place Head TLT Sphere on Phantom Ring

Figure 3-2-2-4



Sphere Aligned with Coil

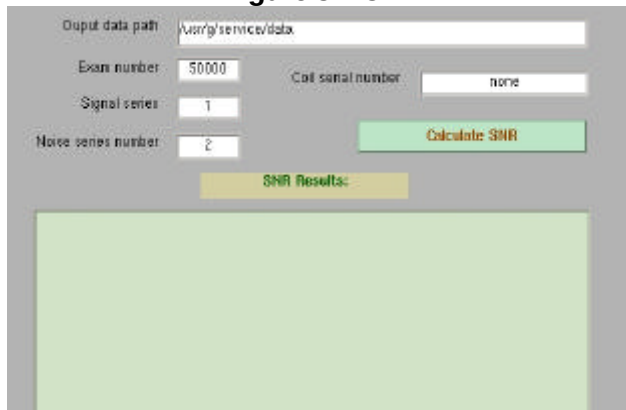
Figure 3-2-2-5



Individual Element ROIs

3-2-3 Data Analysis

1. In the command window enter **snr_for_8hrbrain_QA** and press enter. A screen like that shown in Figure 3-2-3-1 should appear.

Figure 3-2-3-1


Output data path: /usr/g/service/data

Exam number: 50000 Coil serial number: none

Signal series: 1

Noise series number: 2

Calculate SNR

SNR Results:

QA Mode SNR Analysis Screen

2. In the Exam number window enter the exam number as shown in Figures 3-2-3-2 and 3-2-3-3 (The Delete key may be used instead of Backspace).
3. In the Coil serial number window, enter serial number and bay number as shown in Figures 3-2-3-4 and 3-2-3-5 (Optional).

Figure 3-2-3-2


Output data path: /usr/g/service/data

Exam number: 50000 Coil serial number: none

Signal series: 1

Noise series number: 2

Calculate SNR

SNR Results:

Highlight Digits for Exam Number**Figure 3-2-3-3**


Output data path: /usr/g/service/data

Exam number: 51897 Coil serial number: none

Signal series: 1

Noise series number: 2

Calculate SNR

SNR Results:

Type In Exam Number**Figure 3-2-3-4**


Output data path: /usr/g/service/data

Exam number: 51897 Coil serial number: none

Signal series: 1

Noise series number: 2

Calculate SNR

SNR Results:

Highlight Coil Serial Number**Figure 3-2-3-5**


Output data path: /usr/g/service/data

Exam number: 51897 Coil serial number: U7514 b3d

Signal series: 1

Noise series number: 2

Calculate SNR

SNR Results:

Enter Coil Serial Number and Bay Number

4. Click the Calculate SNR button as shown in Figure 3-2-3-6.

Figure 3-2-3-6

Output data path: /usr/g/service/data

Exam number: 51897 Coil serial number: U7514 bay3a

Signal series: 1

Noise series number: 2

Calculate SNR

SNR Results:

Click Calculate SNR Button

Figure 3-2-3-7

Date Path: /usr/g/service/data

Exam number: 50350 Coil serial number: noneP4-1 bay

Signal series number: 1

Noise series number: 2

Calculate SNR

SNR Results:

```
phantom center =(139,139)
Individual Element NEMA SNR from composite images
ROI    signal    noise    snr
1      1384.1    16.07   121.8
2      1449.1    15.97   128.3
3      1514.1    15.43   138.8
4      1524.1    15.09   134.0
5      1613.0    16.77   135.1
6      1682.1    16.51   144.1
7      1610.8    16.58   137.4
8      1478.0    17.05   122.6
mean   1532.0    16.31   132.9
stdev   98.2      0.52    7.9
stdev%  6.4        3.16    6.0
Composite SNR Using NEMA Method
ROI    signal    noise    snr
C      1473.3    16.63   119.8
serial number noneP4-1 bay
```

5. Results will appear as shown in Figure 3-2-3-7.
6. Table 3-2-3-1 shows the test results that correspond to each test. For further explanation, refer to Section 3-2-4, Plotting Tool Procedure.

QA Mode SNR Results

EXAMPLE OUTPUT OF TOOL - TABLE 3-2-3-1

phantom center =(131,137)			
Individual Element NEMA SNR from composite images			
ROI	signal	noise	snr
1	1835.0	19.39	133.8
2	2054.6	19.61	148.1
3	2117.0	19.44	154.0
4	2005.0	19.79	143.3
5	1994.5	19.49	144.7
6	2080.0	19.70	149.3
7	2029.9	20.24	141.8
8	1848.8	20.19	129.5
mean	1995.6	19.73	143.1
stdev	102.8	0.33	8.1
stdev%	5.2	1.66	5.7
Composite SNR Using NEMA Method			
ROI	signal	noise	snr
C	1920.1	19.99	129.8
serial number U7514 bay3a			

Using ROIs shown in Figure 3-2-2-5 signal is measured on each of the two composite images and averaged. The same ROI for each element is used to measure noise on the one filtered difference image

Composite image results

Note: The results are also stored in a file in the /usr/g/service/data directory. The filename will be similar to 8hrbrain_51897_c7ga0522.qa where 51897 is the exam number.

7. Record the coil serial number and the associated filename in Table 3-2-3-2 for future reference.

Note: The analysis tool can be used to process multiple runs by entering a new exam number and coil serial number and pressing the Calculate SNR button again. It is not necessary to exit the analysis tool after each run.

8. Using Table 3-2-3-1 as a reference, compare the test results to the values in Tables 3-2-3-3 and 3-2-3-4. For additional clarification, proceed to Section 3-2-4, Plotting Tool Procedure.

COIL SERIAL NUMBER AND FILENAME - TABLE 3-2-3-2

Date	Coil Serial Number	Filename

COMPOSITE SNR ACCEPTANCE LIMITS – TABLE 3-2-3-3

Nominal	Lower Limit	Upper Limit
131.3	118.2	144.5

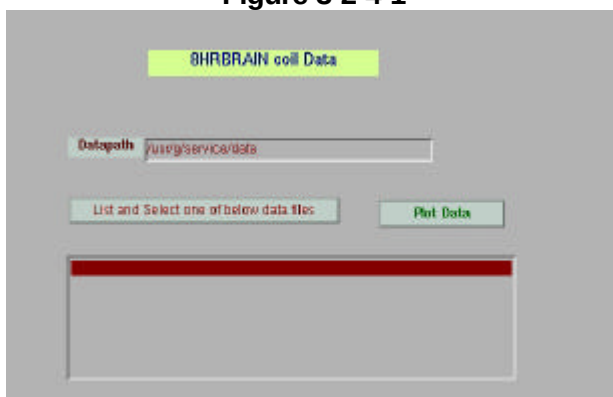
INDIVIDUAL ELEMENT SNR LIMITS - TABLE 3-2-3-4

	element	nominal	lower limit	upper limit
individual elements from composite image	1	126.6	107.6	145.6
	2	143.2	121.7	164.6
	3	150.1	127.6	172.6
	4	152.2	129.4	175.1
	5	154.2	131.0	177.3
	6	155.8	132.5	179.2
	7	149.0	126.6	171.3
	8	130.3	110.8	149.9
	mean	145.2	123.4	166.9

3-2-4 Plotting Tool Procedure

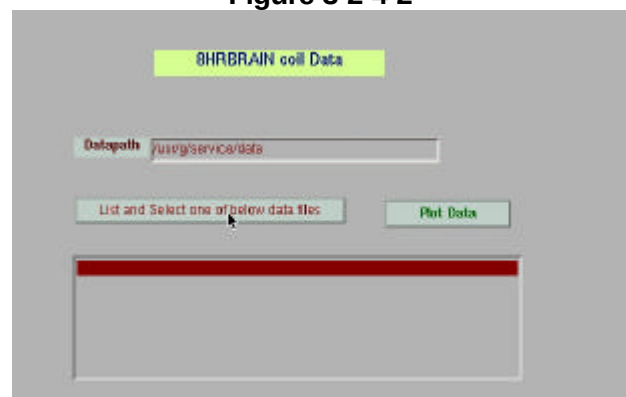
1. Open a command window on the tools desktop.
2. Navigate to the /usr/g/service/cclass directory.
3. Type the command **plot_8hrbrain_data** and press the enter key.
4. The screen shown in Figure 3-2-4-1 will appear.
5. The path to data files, /usr/g/service/data, should be selected by default. If it is not, highlight the text in the type in the correct path.
6. Click on the “List and Select one of below data files” button as shown in Figure 3-2-4-2.
7. Click on the name of the file to be plotted as shown in Figure 3-2-4-3.
8. Click on the Plot Data button as shown in Figure 3-2-4-4 to process the file.
9. Example results for a nominal coil are shown in Figure 3-2-4-5. Coil with poor SNR in Figure 3-2-4-6. Coil with shorted blocking diode in Figure 3-2-4-7. It is interesting to note that the coil whose results are shown in Figure 3-2-4-7 marginally passes the composite image SNR limits but that element 3 performance in individual element results is severely out of normal range.

Figure 3-2-4-1



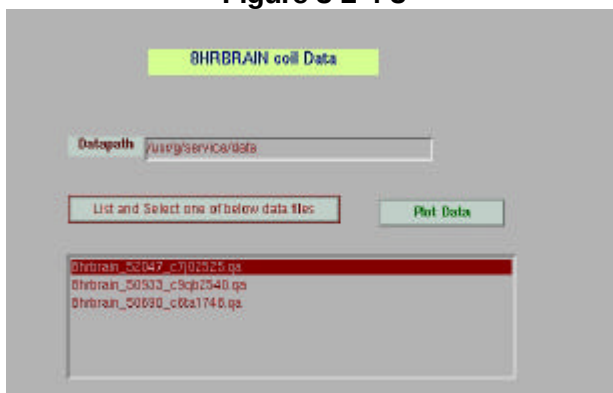
Enter Datapath

Figure 3-2-4-2



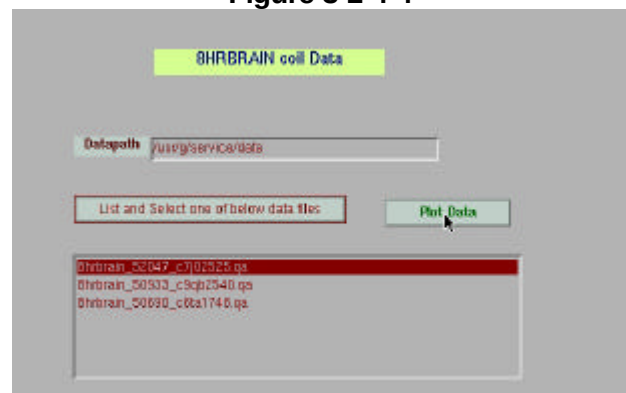
List Files

Figure 3-2-4-3



Select File

Figure 3-2-4-4



Process File

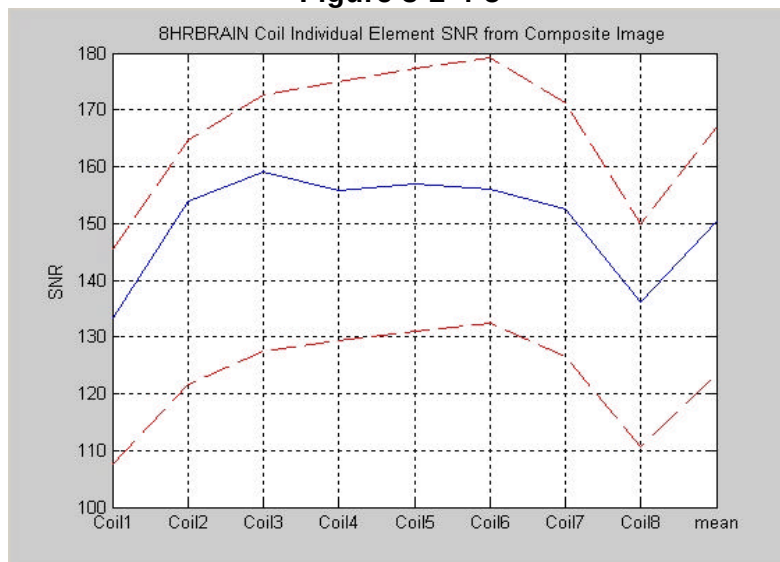
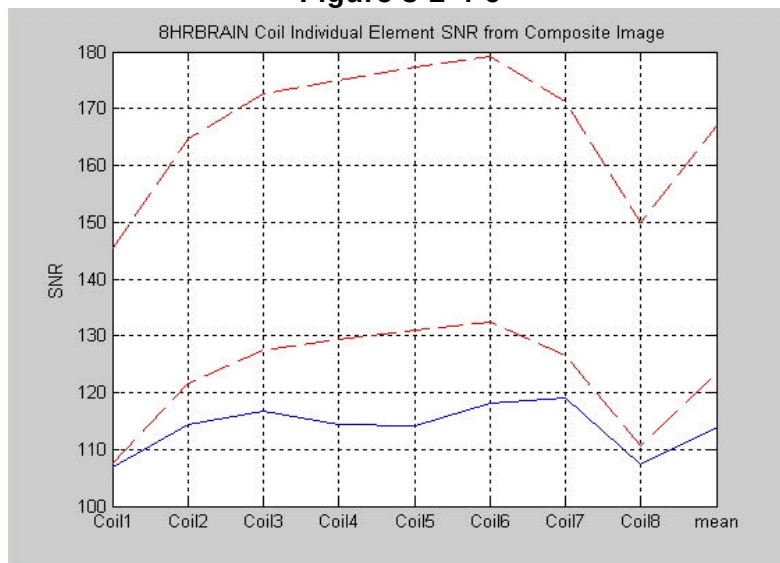
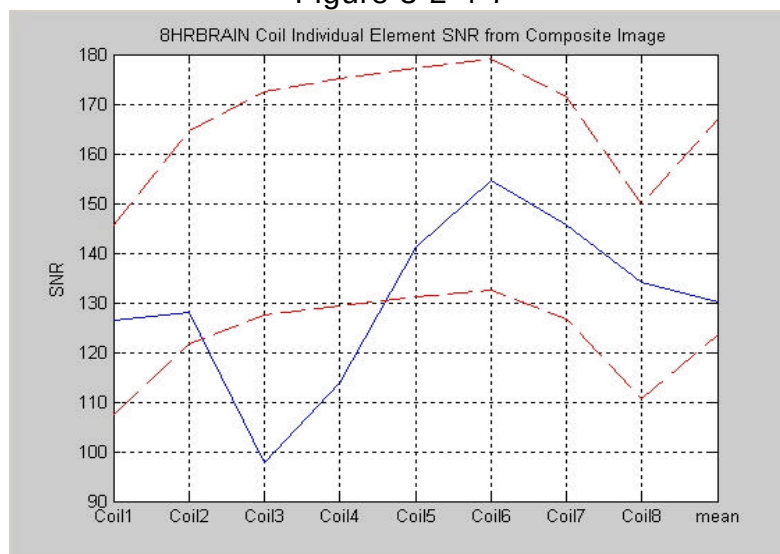
Figure 3-2-4-5**Nominal Individual Element Results****Figure 3-2-4-6****Low SNR Individual Element Results**

Figure 3-2-4-7



Shorted Blocking Diode Individual Element Results

3-2-5 Test Explanation and Results Interpretation

The 8 channel hi res brain coil is a moderately complex phased array. The Quality Assurance tool acquires data looking at the composite image (like the customer sees). Individual element performance is also evaluated using the composite image.

Signal images are acquired using a FSE-XL sequence with CV saveinter set to 1 to save intermediate images into the image database. The scan is taken twice for a total of 18 images. Next, a no-RF noise scan is taken, also with saveinter on for a total of 9 noise images. Individual element signal and noise images no longer yield results that are displayed to the user.

The test employs two types of image analyses: Separate signal and noise images are evaluated for individual element signal, noise and SNR, much like other automated surface coil tests such as those used for both types of 1.5T CTL coils. In addition, a filtered version of the NEMA SNR measurement method is also used. The filter takes out adverse effects of subtle ghosting on noise measured in difference images. Both types use the same individual element regions of interest (ROIs) which are shown in Figure 3-2-2-5. The Composite SNR using NEMA method also uses the ROIs shown in Figure 3-2-2-5. Only results from composite images are displayed to the user.

- Composite SNR acceptance limits are shown in Table 3-2-3-3.
- Example output of the tool is shown in Table 3-2-3-1.

An automated plotting tool is available to evaluate a particular result file against the individual element acceptance limits shown in Table 3-2-3-4. See Section 3-2-4, Plotting Tool Procedure, for instructions. Example plots from good and defective coils are shown in that section.

Chapter 3 Using the 8 Channel High Resolution Brain Array

Positioning the Brain Array on the Patient Cradle

The 8 Channel High Resolution Brain Array is designed for imaging of the entire head.

The array must be placed on the patient cradle with the cable into the magnet.

The 8 Channel High Resolution Brain Array is designed for head first exams only!

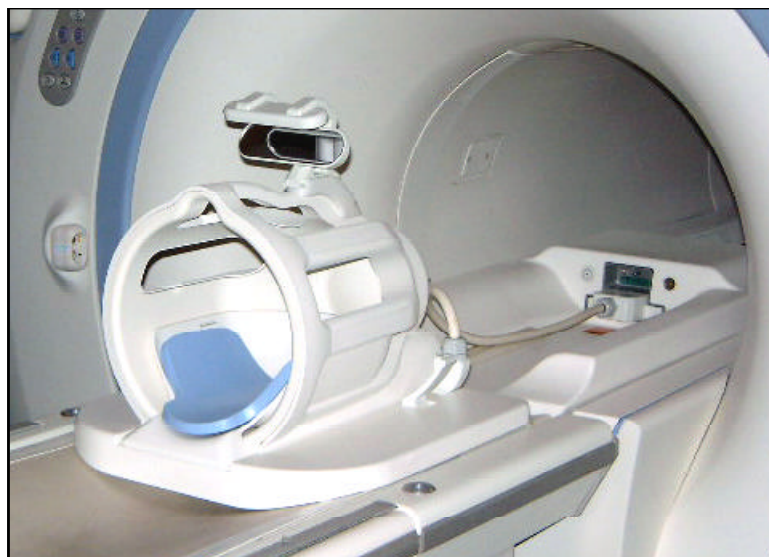
The Brain Array base is designed to rest directly on the patient cradle. Standard GE patient cradle pads fit flush with the end of the Brain Array base. Insert the phased array coil connector into the phased array system port.

(Insert the tabs on the base into the slots on the system patient cradle)



The patient comfort pad should be centered on the head holder. The pad serves to provide additional patient comfort during the scan, helps center the patient in the coil, and is recommended for optimal coil performance.

(Base shown seated on patient cradle, ready for patient positioning)



Positioning the Patient

Place the patient's head on the head holder.



Slide the coil over the patient's head until it touches the shoulders. If necessary, insert wedge pads to help immobilize the patient's head.



Press down the locking levers to prevent the coil from moving during scanning.



Adjusting the Patient Mirror

The mirror is positioned by sliding in the superior or inferior direction until the patient has a clear view of the control room.



Connecting the Cable

Connect the coil to the system using the special phased array 8 channel coil connector.



Establishing the Patient LANDMARK

Use the windows on the coil housing to ensure the patient is centered properly. Landmark on the anatomy of interest and advance the patient into magnet for scanning.

Field of View and Coverage

For head studies a FOV of 24 cm. is suggested, depending upon head size. The 8 Channel High Resolution Brain Array is 24 cm. in diameter. A smaller FOV may be used if desired.

Chapter 4 - Scanning

Selecting the Coil

The coil code that is available for use with the 8 Channel High Resolution Brain Array is **8HRBRAIN**. This coil code uses multiple elements of the coil in an 8 channel array configuration.

Auto Pre-scan

Generally, the Signa Infinity with Excite Technology System auto pre-scan will function normally when used with the 8 Channel High resolution Brain Array. If auto pre-scan returns the message "auto pre-scan failed to complete normally," then manual pre-scan may be required to acquire the image. Consult your GE Signa Infinity with Excite Technology operator's manual for manual pre-scan procedures.

Localizing with the Body Coil

The Signa Infinity with Excite Technology body coil may be used at any time while the 8 Channel High Resolution Brain Array is in the scanner. This allows a large FOV (40 cm.) body coil localizer to be performed, if desired.

IMPORTANT! Using Autoshim

AUTOSHIM is a feature of the software AUTOPRESCAN to improve image quality. It does this by improving the magnetic field homogeneity within the FOV selected. The improvement in image quality is often dramatic when the selected FOV is far off center, and when acquiring FSE or T2 weighted scans, or FAT SAT images.

Magnetization Transfer Techniques

Magnetization transfer techniques are available with the GE Signa Angio software.

Protocols

Refer to the GE protocols on your system.

Chapter 5 - Installation and Maintenance

Installation and Configuration

Coil installation and configuration requires the services of your GE Service Representative.

Please refer to the Service Manual for coil installation and configuration information.

Cleaning

The 8 Channel High Resolution Brain Array, base and pads **must** be cleaned and stored using the following procedure:

- Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.
- **Do not pour any cleaning solution directly on the coil!**
- Let the coil housing and pads dry before use.
- Store the coil in an air-conditioned scan room or equipment room.
- **Under no circumstances should the coil be placed into any type of sterilizer.**

Storage

Store the coil in an air-conditioned scan room or equipment room.

To store the coil, a storage space of 38.1 cm/15.0 in (width) x 46.0 cm/18.1 in (depth) x 35.6 cm/14.0 in (height) is required.

Replaceable Accessories

The following table lists coil accessories that may be purchased for replacement:

Description	GE Part #	MRIDC #
Thick Head Pad	2317112-9	102985
Thin Head Pad	2317112-10	102984
Spacer Pad	2317112-7	101622
Wedge Pad	2317115-8	101407

For additional information, contact:

GE Medical Systems Americas
 USA 800-558-5102
 Canada 800-668-0732

GE Medical Systems Asia
 China 86-21-62192228
 Taiwan 886-2-2505-7900
 Singapore 65-291-8528
 Australia 61-2-9975-5501
 Japan 81-120-48-2630
 Korea 82-31-740-6119
 India 91-80-845-2923

GE Medical Systems Europe
 (33) 1-41-19-76-76

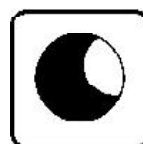
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